

2. Pairing, Products, and Relations

EP2.36 There is a bijection $F : \mathcal{P}(X) \rightarrow \{0, 1\}^X$.

$$F(a)(x) = \begin{cases} 0 & \text{if } x \in A \\ 1 & \text{if } x \notin A \end{cases} \quad \text{which is } 1 - 1 \text{ and onto.}$$

D2.37 Cartesian Product Let F be a function with $\text{dom}(F)$ as a set. $\prod F = \{f : f \text{ is a function} \wedge \text{dom}(f) = \text{dom}(F) \wedge \forall x \in \text{dom}(F) [f(x) \in F(x)]\}$. If $F = \langle A_i : i \in I \rangle$, then $\prod F = \prod_{i \in I} A_i = \{f : f \text{ is a function} \wedge \text{dom}(f) = I \wedge \forall i \in I [f(i) \in A_i]\}$

A2.38 Axiom of Choice If $\langle A_i : i \in I \rangle$ is a sequence of sets such that $\forall i \in I [A_i \neq \emptyset]$, then $\prod_{i \in I} A_i \neq \emptyset$

4. The Natural Numbers

D4.17 Extenders Let $\mathbf{FN} = \{\sigma : \sigma \text{ is a function} \wedge \exists n \in \mathbb{N} [\text{dom}(\sigma) = n]\}$ be the proper class of all functions whose domain is some natural number. An extender is a function $\mathbf{E} : \mathbf{FN} \rightarrow \mathbf{V}$. When you input $\sigma = \{\langle 0, \sigma(0) \rangle, \dots, \langle n, \sigma(n) \rangle\}$ into $\mathbf{E}$, $\mathbf{E}(\sigma)$ outputs the next value $\sigma(S(n))$.

D2.45 Addition

- Define $\langle f_m : m \in \mathbb{N} \rangle$ such that $f_m : \mathbb{N} \rightarrow \mathbb{N}$ is the unique function such that $f_m(0) = m$ and $\forall n \in \mathbb{N} [f_m(S(n)) = S(f_m(n))]$
- In other words, define the extender $\mathbf{E} : \mathbf{FN} \rightarrow \mathbf{V}$ as follows. For any $\sigma \in \mathbf{FN}$,
$$\mathbf{E}(\sigma) = \begin{cases} m & \text{if } \text{dom}(\sigma) = 0 \\ S(\sigma(\bigcup \text{dom}(\sigma))) & \text{if } \text{dom}(\sigma) \neq 0 \end{cases}$$
- $f_m : \mathbb{N} \rightarrow \mathbf{V}$ is the unique function satisfying $\forall n \in \mathbb{N} [f_m(n) = \mathbf{E}(f_m \upharpoonright n)]$.
- Then $m + n = f_m(n)$, and $m + S(n) = (m + n) + 1$.

5. Comparing Sizes of Sets

D5.1 Equinumerosity $A \approx B$ if there exists $f : A \rightarrow B$ which is both $1 - 1$ and onto.

F5.2 $\mathcal{P}(A) \approx \{0, 1\}^A$

D5.4 $A \lesssim B$ means there exists $f : A \rightarrow B$ which is $1 - 1$ and B is at least as big as A . If $A \lesssim B$ but $A \not\approx B$, then $A \lessapprox B$. It is not possible to find $g : A \rightarrow B$ that is both $1 - 1$ and onto. B is strictly bigger in size than A .

L5.5 If $f : A \rightarrow B$ and $g : B \rightarrow C$ are $1 - 1$ functions then $g \circ f : A \rightarrow C$ is $1 - 1$.

L5.6 For sets A, B, C

- $A \lesssim A$
- If $A \lesssim B$ and $B \lesssim C$ then $A \lesssim C$
- If $A \approx B$ and $B \approx C$ then $A \approx C$

T5.7 Cantor For any set X , $X \lessapprox \mathcal{P}(X)$.

5.2 The Schröder Bernstein Theorem

T5.11 Schröder-Bernstein For any sets A and B , if $A \lesssim B$ and $B \lesssim A$, then $A \approx B$.

E5.12 Suppose $f : X \rightarrow Y$ is a $1 - 1$ function. For any $Z \subseteq X$, $Z \approx \text{Im}_f(Z)$.

E5.13 Suppose $I \subseteq A$ and $J \subseteq B$. If $I \approx J$ and $(A \setminus I) \approx (B \setminus J)$, then $A \approx B$.

E5.14 If $n \in \mathbb{N}$ and $A \approx S(n)$, then $\forall a \in A, (A \setminus \{a\} \approx n)$.

E5.15 If $n \in \mathbb{N}$ and $A \approx n$, then if $a \notin A$, $(A \cup \{a\}) \approx S(n)$.

E5.16 Let $n, m \in \mathbb{N}$. Then

- If $f : n \rightarrow n$ is $1 - 1$, then f is onto. There is no $1 - 1$ function from $S(n)$ to n .
- If $m \in n$, then $m \lessapprox n$.
- If $x \subseteq n$, then $x \lessapprox n$.
- $n \lessapprox \mathbb{N}$
- If $A \approx n$, $B \approx m$, and $A \cap B = \emptyset$, then $(A \cup B) \approx (n + m)$.

D5.19 A set is finite if there exists $n \in \mathbb{N}$ such that $n \approx A$. A is infinite if it is not finite. A is countable if $A \lesssim \mathbb{N}$. A is uncountable if it is not countable.

L5.20 If $f : A \rightarrow B$ is a $1 - 1$ function, then for any $X, Y \subseteq A$, if $\text{Im}_f(X) = \text{Im}_f(Y)$, then $X = Y$.

L5.21 For sets A, B, C, D

- If $A \lesssim B$ then $\mathcal{P}(A) \lesssim \mathcal{P}(B)$
- If $A \lessapprox B$ then $A^C \lesssim B^C$
- If $A \lesssim B, C \lesssim D$, and $B \cap D = \emptyset$, then $A \cup C \lesssim B \cup D$

L5.22 If $n \in \mathbb{N}$ and $A \lesssim n$, then A is finite.

L5.23 If $n \in \mathbb{N}$ and there exists an onto function $\sigma : n \rightarrow A$, then $A \lesssim n$

L5.24 If A and B are finite, then so is $A \cup B$.

T5.25 If A is a finite set and f is a function with $\text{dom}(f) = A$ then

- If $X \subseteq A$, then $X \lessapprox A$
- $\text{ran}(f)$ is finite and $\text{ran}(f) \lesssim A$
- If $\forall a \in A [a \text{ is finite}]$ then $\bigcup A$ is finite
- $\mathcal{P}(A)$ is finite

E5.26 If $A \subseteq \mathbb{N}$ is finite and nonempty, $\max(A) = \bigcup A$

E5.27 If $A \lesssim C$ and $B \lesssim D$, then $A \times B \lesssim C \times D$. If A and B are finite, $A \times B$ and A^B are finite.

E5.28 If I is a finite set and $\langle A_i : i \in I \rangle$ is a sequence of sets such that $\forall i \in I [A_i \text{ is finite}]$, then $\prod_{i \in I} A_i$ is finite.

E5.30 For any function, $\text{dom}(f) \approx f$.

6. Orders

Quasi, Partial, Linear, and Well-Orders

D6.2 Quasi Order Reflexive, Transitive

- $\forall x \in X [x \leq x]$
- $\forall x, y, z \in X [(x \leq y \wedge y \leq z) \Rightarrow x \leq z]$

D6.4 Partial Order Irreflexive, Transitive

- $\forall x \in X [x \not\leq x]$
- $\forall x, y, z \in X [(x < y \wedge y < z) \Rightarrow x < z]$

D6.5 Linear Order Irreflexive, Transitive, Comparable

- $\forall x \in X [x \triangleleft x]$
- $\forall x, y, z \in X [(x \triangleleft y \wedge y \triangleleft z) \Rightarrow x \triangleleft z]$
- $\forall x, y \in X [x = y \vee x \triangleleft y \vee y \triangleleft x]$

F6.6 Suppose $\langle X, < \rangle$ is a partial order. Define a relation $\leq$ on X by $x \leq y$ iff $x < y$ or $x = y$. Then $\langle X, \leq \rangle$ is a quasi order where $\forall x, y \in X [(x \leq y \wedge y \leq x) \Rightarrow x = y]$.

C6.8 If $\langle X, < \rangle$ is a partial order and $Y \subseteq X$ then $(Y \times Y) \cap < \rangle$ is a partial order on Y , as a shorthand for $\langle Y, ((Y \times Y) \cap < \rangle)$. Restricted to Y then Z is the same as restricting directly to Z .

D6.9 Maximal / Minimal Element $x \in X$ is maximal if $\forall y \in X [x \not\leq y]$. $x \in X$ is minimal if $\forall y \in X [y \not\leq x]$. There could be multiple in a partial order.

L6.10 A finite non-empty partial order has both a maximal and minimal element.

D6.11 $C \subseteq X$ is a chain if $\forall x, y \in C [x \text{ and } y \text{ are comparable}]$. $A \subseteq X$ is an antichain if $\forall x, y \in A [x \neq y \Rightarrow x \text{ and } y \text{ are incomparable}]$. A chain is maximal if there is no chain $C' \subseteq X$ where $C \subsetneq C'$. $\emptyset$ and singletons are chains and antichains.

L6.12 For a finite partial order, every chain or antichain is contained in a maximal chain or antichain.

D6.13 Well-Order Every non-empty subset has a minimal element.

$\forall A \subseteq X [A \neq \emptyset \Rightarrow \exists a \in A \forall a' \in A [a \leq a']]$.

L6.15 (AC) A linear order $\langle X, < \rangle$ is a well-order iff there is no $f : \mathbb{N} \rightarrow X$ where $\forall n \in \mathbb{N} [f(n) > f(n + 1)]$.

D6.16 For a linear order $\langle X, < \rangle$ $\text{pred}_{\langle X, < \rangle}(x) = \{x' \in X : x' < x\}$, or the set of predecessors of x in X for the ordering $< \rangle$. A subset $A \subseteq X$ is downwards closed if $\forall a \in A \forall x \in X [x < a \Rightarrow x \in A]$. The predecessor subset is downwards closed along with the entire set.

F6.17 For a linear order $\langle X, < \rangle$, if $A \subseteq X$ is downwards closed,

$$\forall a \in A \text{ pred}_{\langle A, < \rangle}(a) = \text{pred}_{\langle X, < \rangle}(a).$$

F6.19 Let $\langle X, < \rangle$ be a well-order and A a downwards closed subset of X . Either $A = X$ or $\exists x \in X$ such that $A = \text{pred}_{\langle X, < \rangle}(x)$.

E6.20 If $\langle X, < \rangle$ is a well-order and $A \subseteq X$, $\langle A, < \rangle$ is a well-order.

E6.21 Let $\langle X, < \rangle$ be a linear order. $f : X \rightarrow X$ is expansive if

$\forall x \in X [f(x) \geq x]$, and order-preserving if

$\forall x, y \in X [x < y \Rightarrow f(x) < f(y)]$. If $\langle X, < \rangle$ is a well-order, every order-preserving f is expansive.

New orders from old

L6.23 Suppose X is a set and $\langle Y, \triangleleft \rangle$ and $\langle Z, \triangleleft \rangle$ are partial orders. Suppose $f : X \rightarrow Y$ and $g : X \rightarrow Z$ are functions. Define $<$ on X by stipulating that $\forall x, x' \in X, x < x' \Leftrightarrow (f(x) \triangleleft f(x') \text{ or } (f(x) = f(x') \text{ and } g(x) \triangleleft g(x')))$. Then,

- $<$ is a partial order on X
- if $\langle Y, \triangleleft \rangle$ and $\langle Z, \triangleleft \rangle$ are both linear orders and $\forall x, x' \in X [(f(x) = f(x') \text{ and } g(x) = g(x')) \Rightarrow x = x']$ then $<$ is also a linear order on X
- similarly for well-orders

C6.24 Let X be a set and $\langle Y, \triangleleft \rangle$ be a partial order. Suppose $f : X \rightarrow Y$ is a function. Define $<^*$ on X by $\forall x, x' \in X, x <^* x' \Leftrightarrow f(x) \triangleleft f(x')$. Then $<^*$ is a partial order on X . If f is $1 - 1$ and $\triangleleft$ is a linear order on Y , then $<^*$ is a linear order on X . Similarly for well-orders. Use $Y = Z, f = g, \triangleleft = \triangleleft$.

EP 6.25 Lexographic / Dictionary Order Define $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ and $g : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ by setting $f(\langle m, n \rangle) = m$ and $g(\langle m, n \rangle) = n$ for all $\langle m, n \rangle \in \mathbb{N} \times \mathbb{N}$. By L6.23, the relation $<_{lex}$ defined on $\mathbb{N} \times \mathbb{N}$ by $\langle m, n \rangle <_{lex} \langle k, l \rangle \Leftrightarrow (f(\langle m, n \rangle) \in f(\langle k, l \rangle) \text{ or } (f(\langle m, n \rangle) = f(\langle k, l \rangle) \text{ and } g(\langle m, n \rangle) \in g(\langle k, l \rangle))) \Leftrightarrow (m \in k \text{ or } (m = k \text{ and } n \in l))$ is a well-order on $\mathbb{N} \times \mathbb{N}$.

D6.26 Suppose $\langle I, < \rangle$ is a well-order and X is a set. For $f, g \in X^I$, if $f \neq g$, define $\Delta(f, g) = \min(\{i \in I : f(i) \neq g(i)\}, < \rangle)$. It is well-defined for the conditions mentioned above.

L6.27 Suppose $\langle X, \triangleleft \rangle$ is a linear order and $\langle I, < \rangle$ is a well-order. Define $\triangleleft$ on $X^I \forall f, g \in X^I$ by $f \triangleleft g \Leftrightarrow (f \neq g \text{ and } f(\Delta(f, g)) \triangleleft g(\Delta(f, h)))$. Then $\triangleleft$ is a linear order on X^I .

D6.28

- $[\mathbb{N}]^n = \{a \in \mathcal{P}(\mathbb{N}) : a \approx n\}$, or subsets of the naturals equinumerous to n .
- $\mathbb{N}^n = \{\sigma : \sigma \text{ is a function and } \text{dom}(\sigma) = n \wedge \text{ran}(\sigma) \subseteq \mathbb{N}\}$, or functions with domain n and range as a subset of the naturals.
- $[\mathbb{N}]^{<\omega} = \{a \in \mathcal{P}(\mathbb{N}) : a \text{ is finite}\}$, or subsets of the naturals equinumerous to finite n .
- $\mathbb{N}^{<\omega} = \{\sigma : \sigma \text{ is a function and } \text{dom}(\sigma) \in \mathbb{N} \wedge \text{ran}(\sigma) \subseteq \mathbb{N}\}$, or functions with a finite domain and range as a subset of naturals.
- $[\mathbb{N}]^{<\omega} = \bigcup_{n \in \mathbb{N}} [\mathbb{N}]^n$ and $\mathbb{N}^{<\omega} = \bigcup_{n \in \mathbb{N}} \mathbb{N}^n$.

EP6.29 $\langle \mathbb{N}, \in \rangle$ is a well-order. Define a linear order $<_{lex}$ on $\mathbb{N}^{\mathbb{N}}$ using L6.27. Since $2^{\mathbb{N}} \subseteq \mathbb{N}^{\mathbb{N}}, \langle 2^{\mathbb{N}}, <_{lex} \rangle$ is also a linear order. Define $F : \mathbb{N}^{<\omega} \rightarrow \mathbb{N}^{\mathbb{N}}$ as follows. $\sigma \in \mathbb{N}^{<\omega}, \text{dom}(\sigma) \in \mathbb{N}$ and $\sigma : \text{dom}(\sigma) \rightarrow \mathbb{N}$. Define $F(\sigma) : \mathbb{N} \rightarrow \mathbb{N}$

$$\text{by } F(\sigma)(n) = \begin{cases} \sigma(n) & \text{if } n \in \text{dom}(\sigma) \\ \text{dom}(\sigma) & \text{if } n \notin \text{dom}(\sigma) \end{cases} \quad \text{for every } n \in \mathbb{N}. F \text{ is } 1 - 1.$$

Since $<_{lex}$ is a linear order on $\mathbb{N}^{\mathbb{N}}$, by C6.24, $<_{lex}$ defined on $\mathbb{N}^{<\omega}$ by $\sigma <_{lex} \tau \leftrightarrow F(\sigma) <_{lex} F(\tau)$ is also a linear order. From EP2.36, we have a $1 - 1$ and onto $F : \mathcal{P}(\mathbb{N}) \rightarrow 2^{\mathbb{N}}$. We can induce $A <_{lex} B \Leftrightarrow F(A) <_{lex} F(B)$ for $A, B \in \mathcal{P}(\mathbb{N})$. Since $[\mathbb{N}]^{<\omega} \subseteq \mathcal{P}(\mathbb{N})$, $\langle [\mathbb{N}]^{<\omega}, <_{lex} \rangle$ is also a linear order.

E6.30 Let $\langle X, < \rangle$ and $\langle Y, \triangleleft \rangle$ be partial orders. Define $\langle X \times Y, \triangleleft \rangle$ by $\langle x, y \rangle \triangleleft \langle x', y' \rangle \Leftrightarrow (x < x' \wedge y \triangleleft y')$. This is a partial order.

E6.31 A linear order $\langle X, < \rangle$ is dense if $\forall x, z \in X$ with $x < z, \exists y \in X$ such that $x < y < z$. $\langle \mathbb{N}^{<\omega}, <_{lex} \rangle$ is dense while $\langle 2^{\mathbb{N}}, <_{lex} \rangle$ is not dense.

E6.32 $<_{lex}$ on $\mathcal{P}(\mathbb{N})$ is defined by the $1 - 1$ and onto F from EP2.36.

- $\forall A, B \in \mathcal{P}(\mathbb{N}), A <_{lex} B$ iff $A \neq B \wedge \min(A \triangle B) \in A$
- $\langle [\mathbb{N}]^{<\omega} \setminus \{\emptyset\}, <_{lex} \rangle$ does not have any maximal or minimal elements
- $\langle [\mathbb{N}]^{<\omega} \setminus \{\emptyset\}, <_{lex} \rangle$ is dense

6.3 Embeddings and Isomorphisms

D6.33 If $\langle X, \triangleleft \rangle$ and $\langle Y, \prec \rangle$ are linear orders, $f : X \rightarrow Y$ is an isomorphism between them if f is $1 - 1$ and onto and $\forall x, y \in X [x \triangleleft y \Leftrightarrow f(x) \prec f(y)]$. Two linear orders are isomorphic if f exists which is an isomorphism.

L6.34 $\langle X, \triangleleft \rangle$ and $\langle Y, \prec \rangle$ are linear orders. Suppose $f : X \rightarrow Y$ is an onto function such that $\forall x, y \in X [x \triangleleft y \Rightarrow f(x) \prec f(y)]$. f is an isomorphism.

D6.35 $\langle X, < \rangle$ and $\langle Y, \prec \rangle$ are linear orders. $f : X \rightarrow Y$ is an embedding if $\forall x, x' \in X [x < x' \Leftrightarrow f(x) \prec f(x')]$ and f is $1 - 1$. If there exists and embedding f , we say that $\langle X, < \rangle$ embeds into $\langle Y, \prec \rangle$ and $\langle X, < \rangle \hookrightarrow \langle Y, \prec \rangle$. $\langle X, < \rangle$ is isomorphic to $\langle Im_f(X), \prec \rangle$.

F6.36 $\langle X, < \rangle$ and $\langle Y, \prec \rangle$ are linear orders. If $f : X \rightarrow Y$ is a function such that $\forall x, x' \in X [x < x' \Rightarrow f(x) \prec f(x')]$, then f is an embedding.

F6.37 $\langle X, < \rangle$ and $\langle Y, \prec \rangle$ are linear orders. Suppose A and B are downwards closed subsets of X and Y . If $f : A \rightarrow B$ is an isomorphism from $\langle A, < \rangle$ to $\langle B, \prec \rangle$, then $\forall a \in A, f \upharpoonright pred_{\langle X, < \rangle}(a)$ is an isomorphism from $\langle pred_{\langle X, < \rangle}(a), < \rangle$ to $\langle pred_{\langle Y, \prec \rangle}(f(a)), \prec \rangle$.

T6.38 Suppose $\langle X, \triangleleft \rangle$ is a finite linear order. $\exists! n \in \mathbb{N}$ such that $\langle X, \triangleleft \rangle$ is isomorphic to $\langle n, \in \rangle$. This isomorphism is unique.

T6.39 Suppose $\langle X, \triangleleft \rangle$ is an infinite linear order such that $\forall x \in X, pred_{\langle X, \triangleleft \rangle}(x)$ is finite. $\langle X, \triangleleft \rangle$ is isomorphic to $\langle \mathbb{N}, \in \rangle$. The isomorphism from X to $\mathbb{N}$ is unique.

CL6.40 $\forall x, y \in X [x < y \Rightarrow f(x) \in f(y)]$.

CL6.41 $ran(f)$ is a downwards closed subset of $\langle \mathbb{N}, \in \rangle$.

D6.42 A linear order $\langle X, \triangleleft \rangle$ has type omega ω if X is infinite and $\forall x \in X$, $pred_{\langle X, \triangleleft \rangle}(x)$ is finite.

7. Countable and Uncountable Sets

Countable Sets

C7.1 If $X \subseteq \mathbb{N}$ is infinite, $\langle X, \in \rangle$ is isomorphic to $\langle \mathbb{N}, \in \rangle$.

C7.2 If X is infinite and countable, then $X \approx \mathbb{N}$.

T7.3 There exist linear orders of type omega on $\mathbb{N} \times \mathbb{N}$, $[\mathbb{N}]^{<\omega}$, and $\mathbb{N}^{<\omega}$. Define $f : \mathbb{N} \rightarrow X$ to show infinite, then $g : X \rightarrow \mathbb{N}$ and $h : X \rightarrow X$. Define $\prec$ as normal, and use it to show $pred_{\langle X, \prec \rangle}(x)$ is finite.

C7.4

- $\mathbb{N} \times \mathbb{N} \approx \mathbb{N}$, $[\mathbb{N}]^{<\omega} \approx \mathbb{N}$, and $\mathbb{N}^{<\omega} \approx \mathbb{N}$
- $\forall n \in \mathbb{N} [n \geq 1 \Rightarrow \mathbb{N}^n \wedge [\mathbb{N}]^{<\omega} \approx \mathbb{N}]$

L7.5 Suppose that $\langle A_n : n \in \mathbb{N} \rangle$ and $\langle f_n : n \in \mathbb{N} \rangle$ are sequences such that $\forall n \in \mathbb{N}, f_n : A_n \rightarrow \mathbb{N}$ is $1 - 1$. Then $\bigcup_{n \in \mathbb{N}} A_n$ is countable.

L7.6 (AC) A countable union of countable sets is countable. If $\langle A_n : n \in \mathbb{N} \rangle$ is a sequence of countable sets, then $\bigcup_{n \in \mathbb{N}} A_n$ is countable.

L7.8 The set of rationals $\mathbb{Q}$ is countable, i.e. $\mathbb{Q} \rightarrow \mathbb{N}$.

E7.9 Let $\langle X, \triangleleft \rangle$, $\langle Y, \triangleleft \rangle$ and $\langle Z, \triangleleft \rangle$ be linear orders. Suppose $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are isomorphisms. $f^{-1} : Y \rightarrow X$ and $g \circ f : X \rightarrow Z$ also are.

E7.10 Define a sequence of relations $\triangleleft_n : n \in \mathbb{N}$ by induction on $n \in \mathbb{N}$. Let $\triangleleft_0$ be $\emptyset$. Suppose $\triangleleft_n$ is a relation on $[\mathbb{N}]^n$. Define $\triangleleft_{n+1}$ as follows. Fix $a, b \in [\mathbb{N}]^{S(n)}$. If $\max(a) < \max(b)$, $a \triangleleft_{n+1} b$. If $\max(a) = \max(b)$, $a \cap \max(b)$ and $b \cap \max(b)$ are in $[\mathbb{N}]^n$. Then if $a \cap \max(a) \triangleleft_n b \cap \max(b)$, $a \triangleleft_{n+1} b$. Now define $\triangleleft_n : n \in \mathbb{N}$ by induction. $\triangleleft_0 = \emptyset$. Suppose $\triangleleft_n$ is a relation on $\mathbb{N}^n$. Define $\triangleleft_{n+1}$ as follows. Fix $\sigma, \tau \in \mathbb{N}^{n+1}$. $ran(\sigma)$ and $ran(\tau)$ are finite non-empty subsets of $\mathbb{N}$. If $\max(ran(\sigma)) < \max(ran(\tau))$, $\sigma \triangleleft_{n+1} \tau$. $\sigma \upharpoonright n$ and $\tau \upharpoonright n$ are members of $\mathbb{N}^n$. If $\max(ran(\sigma)) = \max(ran(\tau))$ and if $\sigma \upharpoonright n \triangleleft_n \tau \upharpoonright n$, then $\sigma \triangleleft_{n+1} \tau$. If $\max(ran(\sigma)) = \max(ran(\tau))$, $\sigma \upharpoonright n = \tau \upharpoonright n$, and $\sigma(n) < \tau(n)$, then $\sigma \triangleleft_{n+1} \tau$.

- $\triangleleft_n : n \in \mathbb{N}$ is well-defined, $\forall n \in \mathbb{N}$, $\langle [\mathbb{N}]^n, \triangleleft_n \rangle$ is a linear order of type ω .

2. $\triangleleft_n : n \in \mathbb{N}$ is well-defined, $\forall n \in \mathbb{N}$, $\langle \mathbb{N}^n, \triangleleft_n \rangle$ is a linear order of type ω .

3. We can then prove (2) and (3) of T7.3 without AC.

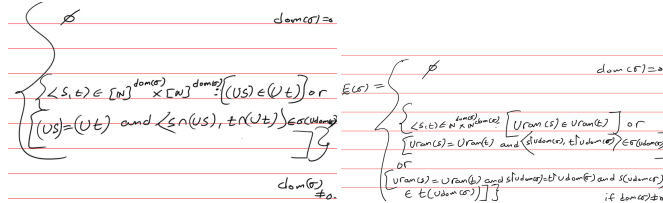
E7.11 Extenders for the following:

- $f(0) = f(1) = 1, f(n) = f(n-1) + f(n-2) \forall n > 1$

$$E(\sigma) = \begin{cases} 1 & dom(\sigma) \in 2 \\ \sigma(\bigcup dom(\sigma)) + \sigma(\bigcup \bigcup dom(\sigma)) & \text{for valid } dom(\sigma) \text{ etc.} \\ \emptyset & \text{otherwise} \end{cases}$$

- $f(n) = \triangleleft_n$ on $[\mathbb{N}]^n$

- $f(n) = \triangleleft_n$ on $\mathbb{N}^n$



7.2 Sets of Size Continuum

F7.12 If $x, y \in \mathbb{R}$ and $x < y$, there is a $q \in \mathbb{Q}$ with $x < q < y$.

L7.13 $2^{\mathbb{N}} \approx \mathbb{R} \approx \mathcal{P}(\mathbb{Q})$

T7.14 These sets are equinumerous: $2^{\mathbb{N}}, \mathbb{N}^{\mathbb{N}}, \mathcal{P}(\mathbb{N} \times \mathbb{N}), \mathcal{P}(\mathbb{N}), \mathcal{P}(\mathbb{Q}), \mathbb{R}$. $2^{\mathbb{N}} \subseteq \mathbb{N}^{\mathbb{N}} \subseteq \mathcal{P}(\mathbb{N} \times \mathbb{N})$. So $2^{\mathbb{N}} \approx \mathbb{N}^{\mathbb{N}} \approx \mathcal{P}(\mathbb{N} \times \mathbb{N})$. $\mathbb{N} \approx \mathbb{N} \times \mathbb{N}$. By L5.22, $\mathcal{P}(\mathbb{N}) \approx \mathcal{P}(\mathbb{N} \times \mathbb{N}) \approx \mathcal{P}(\mathbb{N})$ so $\mathcal{P}(\mathbb{N}) \approx \mathcal{P}(\mathbb{N} \times \mathbb{N})$. By L7.8, $\mathbb{Q} \approx \mathbb{N}$. By F5.2, $2^{\mathbb{N}} \approx \mathbb{R} \approx \mathcal{P}(\mathbb{Q}) \approx \mathcal{P}(\mathbb{N}) \approx 2^{\mathbb{N}}$. We also have $2^{\mathbb{N}} \approx \mathbb{N}^{\mathbb{N}} \approx \mathcal{P}(\mathbb{N} \times \mathbb{N}) \approx 2^{\mathbb{N}}$.

D7.15 A set X has size continuum or size $\mathfrak{c}$ if $X \approx \mathcal{P}(\mathbb{N})$.

L7.16 $(r, s) = \{x \in \mathbb{R} : r < x < s\}$ has size $\mathfrak{c}$. The function

$$f(x) = \begin{cases} \frac{x-t}{s-t} & \text{if } x \geq t \\ \frac{x-t}{x-r} & \text{if } x < t \end{cases} \text{ is well-defined and } 1 - 1 \text{ and onto.}$$

E7.17 Let $l \subseteq \mathbb{R}^2$ be a line. Define $\phi : \mathbb{R} \rightarrow l$ as $\phi(x) = \langle x, mx + c \rangle$ and $\phi(y) = \langle c, y \rangle$ for the different line cases. $l \approx \mathbb{R}$. For $a < b$ and $c < d$, let $m = \frac{d-c}{b-a}$ and $p = c - ma$. Then $f : (a, b) \rightarrow (c, d)$ is $1 - 1$ and onto.

L7.21 If $A \approx B$ and $A \neq \emptyset$, then there exists an onto $g : B \rightarrow A$.

L7.22 (AC) Suppose A and B are sets and $f : B \rightarrow A$ is onto. Then $A \approx B$.

L7.23 Let A, B, C be sets and suppose $f : C \rightarrow B$ is onto. Then $A^B \approx A^C$.

C7.24 If $B \approx C$, then $A^B \approx A^C$.

C7.25 If $A \approx D, B \approx C$, and $B \neq \emptyset$, then $A^B \approx D^C$.

L7.26 There exists a sequence $\langle A_n : n \in \mathbb{N} \rangle$ of pairwise disjoint infinite subsets of $\mathbb{N}$ such that $\bigcup_{n \in \mathbb{N}} A_n = \mathbb{N}$. Define $B_n = \{(n, m) : m \in \mathbb{N}\}$, and let $A_n = f^{-1}(B_n)$.

L7.27 Suppose A, B , and C are sets with $B \cap C = \emptyset$. Then

$A^B \times A^C \approx A^{B \cup C}$. Define $F : A^B \times A^C \rightarrow A^{B \cup C}, F(f, g) = f \cup g$, and show bijectivity.

C7.28 $\mathbb{N}^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}} \approx \mathbb{N}^{\mathbb{N}}$. Define $A \cup B = \mathbb{N}, A \cap B = \emptyset$. Then $\mathbb{N}^A \approx \mathbb{N}^B \approx \mathbb{N}^{\mathbb{N}}$ and $\mathbb{N}^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}} \approx \mathbb{N}^A \times \mathbb{N}^B \approx \mathbb{N}^{A \cup B} \approx \mathbb{N}^{\mathbb{N}}$.

C7.29 $\mathbb{R}^2$ has size $\mathfrak{c}$. This can be used to count lines and planes. Use $\mathbb{R}^2 \approx \mathbb{R}^{\{0\}} \times \mathbb{R}^{\{1\}} \approx \mathbb{R} \times \mathbb{R} \approx \mathbb{N}^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}} \approx \mathbb{N}^{\mathbb{N}} \approx \mathbb{R}$.

EP7.30 Let $\mathcal{L}$ denote the set of lines in $\mathbb{R}^2$. $\mathcal{L} = \mathcal{L}_0 \cup \mathcal{L}_1$ where $\mathcal{L}_0 = \{l : l \text{ satisfies } y = mx + c, m, c, \in \mathbb{R}\}$ and $\mathcal{L}_1 = \{l : l \text{ satisfies } x = c, c, \in \mathbb{R}\}$. Then $\mathcal{L}_0 \approx \mathbb{R}^2 \approx \mathbb{R}$ and $\mathcal{L}_1 \approx \mathbb{R}$, so $\mathcal{L}$ has size $\mathfrak{c}$.

L7.31 Let A, B, C be sets. $A^{(B \times C)} \approx (A^B)^C$. Define $F : A^{(B \times C)} \rightarrow (A^B)^C$ using $F(f) : C \rightarrow A^B$ as $F(f)(c)(b) = f(\langle b, c \rangle)$.

C7.32 $(\mathbb{N}^{\mathbb{N}})^{\mathbb{N}} \approx \mathbb{N}^{\mathbb{N}}$. Use $(\mathbb{N}^{\mathbb{N}})^{\mathbb{N}} \approx \mathbb{N}^{(\mathbb{N} \times \mathbb{N})}$, and $\mathbb{N} \times \mathbb{N} \approx \mathbb{N}$.

C7.33 $\mathbb{R}^{\mathbb{N}}$ has size $\mathfrak{c}$. Use $\mathbb{R} \approx \mathbb{N}^{\mathbb{N}}$.

D7.34 A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous if for each $x \in \mathbb{R}$ and each $\epsilon > 0$, there exists $\delta > 0$ such that $Im_f((x - \delta, x + \delta)) \subseteq (f(x) - \epsilon, f(x) + \epsilon)$. A set $U \subseteq \mathbb{R}$ is an open interval if there exists $r, s \in \mathbb{R}$ such that

$U = (r, s) = \{x \in \mathbb{R} : r < x < s\}$. $U \subseteq \mathbb{R}$ is open if it is the union of a collection of open intervals.

L7.35 There are only $\mathfrak{c}$ many continuous functions from $\mathbb{R}$ to $\mathbb{R}$.

L7.36 There are only $\mathfrak{c}$ many open subsets of $\mathbb{R}$.

E7.37 $\mathbb{R}^{\mathbb{R}} \approx 2^{\mathbb{R}}$. First, $2^{\mathbb{R}} \subseteq \mathbb{R}^{\mathbb{R}}$, so $2^{\mathbb{R}} \approx \mathbb{R}^{\mathbb{R}}$. Next, define $f : \mathbb{R} \rightarrow \mathbb{R}$. $f \in \mathbb{R}^{\mathbb{R}}$, $f \subseteq \mathbb{R} \times \mathbb{R}$, $f \in \mathcal{P}(\mathbb{R} \times \mathbb{R})$. So $\mathbb{R}^{\mathbb{R}} \subseteq \mathcal{P}(\mathbb{R} \times \mathbb{R})$, and $\mathbb{R}^{\mathbb{R}} \approx \mathcal{P}(\mathbb{R} \times \mathbb{R}) \approx 2^{\mathbb{R} \times \mathbb{R}} \approx 2^{\mathbb{R}}$ as $\mathbb{R} \approx \mathbb{R} \times \mathbb{R}$.

E7.38 There are only countably many algebraic real numbers. Almost all real numbers are transcendental. $a \in \mathbb{R}$ is algebraic if there exists a non-zero polynomial $p(X) \in \mathbb{Z}[X]$ such that $p(a) = 0$. If a is not algebraic, it is transcendental.

E7.39 A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is increasing if $\forall x, y \in \mathbb{R} [x \leq y \Rightarrow f(x) \leq f(y)]$. There are only $\mathfrak{c}$ many increasing functions.

E7.40 Let $X \subseteq 2^{\mathbb{N}}$ be countable. Then $(2^{\mathbb{N}} \setminus X) \approx 2^{\mathbb{N}}$. If T is the set of transcendental real numbers, $T \approx \mathbb{R}$. Since $(2^{\mathbb{N}} \setminus X) \subseteq 2^{\mathbb{N}}$, $(2^{\mathbb{N}} \setminus X) \approx 2^{\mathbb{N}}$. We want to show that $2^{\mathbb{N}} \approx (2^{\mathbb{N}} \setminus X)$ so that we can apply Schröder Bernstein. Let $A, B \subseteq \mathbb{N}$ be infinite sets such that $A \cap B = \emptyset$ and $A \cup B = \mathbb{N}$. By C5.33, fix bijections $\psi : A \rightarrow \mathbb{N}$ and $\varphi : B \rightarrow \mathbb{N}$. Define $G : 2^{\mathbb{N}} \rightarrow (2^{\mathbb{N}} \setminus X)$ as follows by defining $G(f) : \mathbb{N} \rightarrow 2$. Let $n \in \mathbb{N}$. If $n \in A$, define $G(f)(n) = f(\psi(n)) \in 2$. If $n \in B$, $e(\varphi(n)) \in X \subseteq 2^{\mathbb{N}}$, and $e(\varphi(n))(n) \in 2$. If $e(\varphi(n))(n) = 0$, then $G(f)(n) = 1$, if $e(\varphi(n))(n) = 1$, then $G(f)(n) = 0$. Since either $n \in A$ or $n \in B$, $G(f)(n) \in 2$ and $G(f) \in 2^{\mathbb{N}}$. If $G(f) \in X$, then there exists $k \in \mathbb{N}$ with $e(k) = G(f)$ and $n \in B$ with $\varphi(n) = k$. But since $n \in B$, $G(f)(n) \neq e(\varphi(n))(n) = e(k)(n) = G(f)(n)$, a contradiction. This shows $G(f) \notin X$. Now we show that G is $1 - 1$. Fix $f \neq f' \in 2^{\mathbb{N}}$. There exists $k \in \mathbb{N}$ with $f(k) \neq f'(k)$. There exists $n \in A$ with $\psi(n) = k$. $G(f)(n) = f(\psi(n)) = f(k) \neq f'(k) = f'(\psi(n)) = G(f')(n)$. Then $G(f) \neq G(f')$. We have shown that $2^{\mathbb{N}} \approx (2^{\mathbb{N}} \setminus X)$ as needed.

8. More about Partial and Linear Orders

8.2 More about Linear Orders

D8.8 Let $\langle X, < \rangle$ be a partial order and $A \subseteq X$. $x \in X$ is an upper bound of A if $\forall a \in A [a \leq x]$. x is a lower bound if $\forall a \in A [x \leq a]$. Let U be the set of upper bounds of A and L be the set of lower bounds of A . If there exists $u \in U$ such that $\forall x \in U [u \leq x]$, then u is the supremum of A in X or $sup_X(A)$ or minimal upper bound. For L and $[x \leq l]$, it is called the infimum or $inf_X(A)$ or greatest lower bound. There can only be at most one supremum or infimum.

EP8.9 Let $A = \{p \in \mathbb{Q} : p^2 < 2\}$. $sup_{\mathbb{Q}}(A)$ and $inf_{\mathbb{Q}}(A)$ do not exist, but $sup_{\mathbb{R}}(A) = \sqrt{2}$ and $inf_{\mathbb{R}}(A) = -\sqrt{2}$.

D8.10 Let $\langle X, < \rangle$ be a linear order. A pair $\langle A, B \rangle$ is a cut of $\langle X, < \rangle$ if A is downwards closed, B is upwards closed, and A and B partition X i.e. $A \cap B = \emptyset$ and $A \cup B = X$.

F8.11 Suppose $\langle X, < \rangle$ is a linear order and $Y \subseteq X$. If $z \in X \setminus Y$, $A = \{a \in Y : a < z\}$, $B = \{b \in Y : z < b\}$, then $\langle A, B \rangle$ is a cut of $\langle Y, < \rangle$.

D8.13 A linear order $\langle X, < \rangle$ is dense if $\forall x, y \in X \exists z \in X [x < y \Rightarrow x < z < y]$.

D8.14 A linear order is without endpoints if it has neither a maximal nor minimal element. A countable dense linear order with no endpoints is universal for all countable linear orders; every countable linear order embeds into such an order.

T8.15 (Cantor, AC) Suppose $\langle X, < \rangle$ is a non-empty dense linear order without endpoints. Let $\langle Y, < \rangle$ be any countable linear order. Then $\langle Y, < \rangle \hookrightarrow \langle X, < \rangle$.

T8.16 (Cantor) Let $\langle X, < \rangle$ and $\langle Y, < \rangle$ be non-empty countable dense linear orders without endpoints. Then they are isomorphic.

E8.17 Embedding is a quasi-order (reflexive, transitive) on linear orders.

E8.19 $\langle X, < \rangle \hookrightarrow \langle Y, < \rangle \wedge \langle Y, < \rangle \hookrightarrow \langle X, < \rangle$ does not imply isomorphism. Take $X = \mathbb{Q} \cap [0, 1]$ and $Y = \mathbb{Q} \cap (0, 1)$.

E8.20 We can have $\langle X, < \rangle \not\hookrightarrow \langle Y, < \rangle \wedge \langle Y, < \rangle \not\hookrightarrow \langle X, < \rangle$ (incomparability).

Take $X = \langle \mathbb{N}, \in \rangle$ and $Y = \langle \mathbb{N}, \ni \rangle$.